

A NON-CAYLEY VERTEX-TRANSITIVE 3-RAINBOW DOMINATION REGULAR GRAPH

ABSTRACT. We exhibit a connected cubic vertex-transitive graph X of order 108 such that $\gamma_{r3}(X) = 54$ and X is not a Cayley graph. The graph is CVT[108, 42] in the cubic vertex-transitive census of Potočník, Spiga, and Verret. Its optimal 3-rainbow dominating function is explicit, and its non-Cayley property follows from an exact computation of its full automorphism group. This answers the non-Cayley existence question in Kuzman's Question (Q5) for $d = 3$.

A *3-rainbow dominating function* on a graph G is a map $f: V(G) \rightarrow 2^{\{1,2,3\}}$ such that

$$f(v) = \emptyset \implies \bigcup_{u \in N(v)} f(u) = \{1, 2, 3\}.$$

Its weight is $w(f) = \sum_{v \in V(G)} |f(v)|$, and the minimum possible weight is the 3-rainbow domination number $\gamma_{r3}(G)$. A cubic graph G is *3-rainbow domination regular*, or *3-RDR*, when $\gamma_{r3}(G) = |V(G)|/2$. Kuzman asked whether a non-Cayley vertex-transitive 3-RDR graph exists [1, Question (Q5)].

Theorem 1. *Let X be the graph specified in Appendix A. Then X is a connected cubic vertex-transitive non-Cayley graph of order 108, and*

$$\gamma_{r3}(X) = 54.$$

Consequently, non-Cayley vertex-transitive 3-RDR graphs exist.

Proof. Decoding the sparse6 string in Appendix A gives a connected cubic graph on 108 vertices. It is isomorphic to the census graph CVT[108, 42] [2, 3].

Set

$$\begin{aligned} C_1 &= \{4, 5, 18, 19, 23, 33, 52, 54, 60, 61, 67, 74, 78, 79, 83, 86, 104, 107\}, \\ C_2 &= \{8, 10, 11, 12, 21, 35, 40, 55, 58, 63, 65, 72, 80, 82, 84, 98, 99, 101\}, \\ C_3 &= \{1, 20, 24, 26, 31, 34, 37, 38, 39, 41, 50, 56, 76, 85, 88, 97, 100, 103\}. \end{aligned}$$

These sets are pairwise disjoint and each has size 18. Every vertex outside $C_1 \cup C_2 \cup C_3$ has exactly one neighbor in each C_i . Hence the assignment

$$f(v) = \begin{cases} \{i\}, & v \in C_i, \\ \emptyset, & v \notin C_1 \cup C_2 \cup C_3 \end{cases}$$

2020 *Mathematics Subject Classification.* 05C69, 05C25.

Key words and phrases. rainbow domination, Cayley graph, vertex-transitive graph, cubic graph.

is a 3-rainbow dominating function of weight 54.

For completeness, let g be any 3-rainbow dominating function on a cubic graph G of order n , let $w = w(g)$, and put $V_0 = \{v : g(v) = \emptyset\}$. Each vertex of V_0 receives all three colors from its neighborhood, while each unit of weight is counted at most three times. Thus

$$3|V_0| \leq \sum_{v \in V_0} \sum_{u \in N(v)} |g(u)| \leq 3w.$$

Also $|V(G) \setminus V_0| \leq w$, so $n \leq 2w$. Therefore $\gamma_{r3}(G) \geq n/2$, and the displayed function proves $\gamma_{r3}(X) = 54$.

It remains to verify the symmetry assertions. An exhaustive enumeration of all adjacency-preserving permutations gives, for $A = \text{Aut}(X)$,

$$|A| = 216, \quad A \text{ is transitive}, \quad |[A, A]| = 54, \quad A/[A, A] \cong C_2 \times C_2.$$

Consequently, A has exactly three subgroups of index 2. Each has two vertex orbits, both of size 54. If X were a Cayley graph, A would contain a regular subgroup of order 108; such a subgroup would have index 2 in A and would be transitive, a contradiction. Hence X is non-Cayley. $\square$

Remark 2. Theorem 1 settles the non-Cayley existence question in Kuzman's Question (Q5) for $d = 3$. No claim is made here about minimum order or the cases $d \geq 4$.

APPENDIX A. THE WITNESS

Concatenate the following four strings without whitespace and decode the result in standard sparse6 format. Number the decoded vertices $0, 1, \dots, 107$ and then add 1 to every label.

```
:~?@k_GAA_WAC'WM@_gWGawIF_wQDb?yNcHEL'g]GaWgPcpOUd']Wc'e
YexUJbGuMGw}OEhw_GiGchY]ggXMSdX]WeQ]jJAymJzApkjMsjYE\fh}
_grIcOIS}hbMvMRqwMbu{Njs~oKFAoz'CpZlNjYijjI{okTHRqcp0qst
PsTJLTKlTmzXhIzeyVz|@VlhZVLp\rK|bqDfIre@cstPfvd|dxEXf
```

REFERENCES

- [1] B. Kuzman, *On cubic rainbow domination regular graphs*, Discrete Appl. Math. **373** (2025), 26–38. doi:10.1016/j.dam.2025.04.046.
- [2] P. Potočník, P. Spiga, and G. Verret, *Cubic vertex-transitive graphs on up to 1280 vertices*, J. Symbolic Comput. **50** (2013), 465–477. doi:10.1016/j.jsc.2012.09.002.
- [3] J. Azarija, K. Berčič, F. Koprivec, T. Marc, P. Potočník, P. Rotar, P. Spiga, G. Verret, and J. Vidali, *Cubic vertex-transitive graphs on up to 1280 vertices*, Zenodo, version 1 (2022). doi:10.5281/zenodo.6576526.